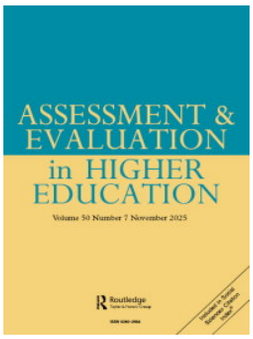

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Interventions and facilitators of oral assessment performance in higher education: a systematic review

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ABSTRACT

This systematic review examines the factors that facilitate student performance in oral assessments at the undergraduate and master's levels across diverse academic disciplines. As higher education increasingly shifts from traditional written assessments to alternative summative methods—partly in response to the rise of generative artificial intelligence (GAI)—understanding the strategies that enhance oral assessment performance has become critical. A comprehensive search of databases, including ERIC, EBSCO Host, Education Full Text, PsycINFO, JSTOR, and Web of Science, identified 24 relevant studies encompassing qualitative, quantitative, and mixed-methods research. A narrative synthesis approach was employed to summarise key findings. The synthesis identified several facilitators of student performance, including structured opportunities for practice, timely and constructive feedback from instructors and peers, the integration of technology (e.g. virtual reality), and the use of self-reflective strategies. These findings highlight practical approaches for educators to enhance student outcomes in oral assessments and suggest avenues for future research to optimise assessment practices in higher education.

KEYWORDS

Oral assessment; higher education; student performance; technology enhanced learning

Introduction

Oral and written communication competencies are crucial for students within a variety of disciplines (Love et al., 1993; Sokhanvar et al., 2021; Wu et al., 2023). Students often get practice developing their written communication skills through assessments such as reports, essays, and proposals (Love et al., 1993; Graves et al., 2010; Tasker, 2022), whereas oral communication is typically assessed using individual or group presentations that are based on open-ended projects (Piirto, 2000; Mandeville et al., 2017). In addition to evaluating communication skills, instructors use these types of assessments to assess a student's conceptual knowledge (Kitto, 2008; Gere et al., 2019; Harsy, 2024) and critical thinking skills (Calma & Davies, 2025).

There are growing concerns about assessment integrity and learning quality due to the emergence of generative AI (GAI) (Sullivan et al., 2023; Chen et al., 2024; Kostanek & Li, 2025). At best, using GAI saves students time, supports their learning through personalised feedback, and can

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lead to less stress in solving open-ended problems (Urban et al., 2024). However, overreliance on GAI can prevent students from developing critical thinking skills and can give students incorrect information (Krupp et al., 2024). In response to concerns raised by GAI, several instructors are replacing traditional written assignments with oral assessments (Moorhouse et al., 2023), as they often require spontaneous, real-time responses (Eachempati et al., 2025).

Findings from oral assessment research can be grouped into three main areas. First, student perceptions and attitudes towards oral examinations (Kelly et al., 2010; Crecelius et al., 2021; Delson et al., 2022; O'Neill, 2024). Generally undergraduate students agree that oral assessments are more useful than written examinations (Kelly et al., 2010). Oral examinations have also shown to encourage students to engage more deeply with the material than they otherwise would (Delson et al., 2022). However, students have also reported that oral-based assessments in particular left them feeling stressed and anxious (Crecelius et al., 2021). Second, research has identified various factors that can either hinder or enhance performance on oral examinations (Cheng & Warren, 2005; Hattie & Timperley, 2007; De Grez et al., 2009; Econopouly et al., 2010; Kolber, 2011; Van Ginkel et al., 2015). A systematic literature review by Van Ginkel et al. (2015) categorised these factors into seven key course design principles aimed at supporting student learning and success in oral assessments. Third, using oral-based assessments are useful for helping students apply their understanding of course material. For example, Nelson (2010) found that Calculus I students who did an oral examination before each written examination performed better on the final examination than those who did not choose to do the oral examination. Additionally, students who engaged in the oral examinations were retained at the university and passed the subsequent course, Calculus II, at higher rates that those who didn't do the oral examinations. Performance on oral examinations can also predict performance across assessment types. Kim et al. (2022) found that students' scores on their oral and written examinations midterms more accurately predicted performance on a written final examination than the students' score on the written midterm examination alone.

There is also evidence that oral examinations more accurately and more deeply probe students' understanding. Sato et al. (2019) compared scores on written examinations to student understanding expressed during retrospective interviews. Over one quarter of the participants showed discrepancies between performance in the written examination and understanding demonstrated in the oral interview with students showing more nuanced understanding in their interview answers.

Over the last decade, research has increasingly explored the role of technology in supporting students' preparation for and performance in oral assessments (Belboukhaddaoudi and Van Ginkel et al., 2019; Davis et al., 2020). Two key areas have been identified: the use of virtual reality (VR) to provide practice opportunities and feedback, and the use of video recordings for self-reflection and remote assessment. However, findings on the effectiveness of these methods remain mixed, particularly regarding their impact on students' presentation anxiety (Boetje et al., 2020; Davis et al., 2020).

Given the importance of developing oral communication skills in multiple disciplines and emerging findings regarding technology's use in oral-based assessments, our systematic literature review will address the following research question: *What interventions facilitate university students' performance in oral assessments*?

Method

Search terms

Search terms were developed through discussions among the authors, initial scoping searches, and relevant literature known to the authors. In addition, keywords used by Van Ginkel et al. (2015) in their systematic literature review on principles for developing oral presentation competence were drawn on. Keywords were intentionally selected to be broad, particularly for the type

of intervention, to capture any method aimed at improving oral assessment performance. However, the search was designed to focus specifically on studies involving a university population and related to some form of oral assessment. Terms were adjusted based on the specific requirements of each database, and relevant truncations were applied.

Universit OR Higher Education OR Undergrad* OR Postgrad* AND Oral assessment OR Oral exam* OR Oral Presentation AND Perform* OR Impact* OR Competence* OR Improve* OR view*.*

Inclusion criteria

Inclusion and exclusion criteria were developed within the 'SPIDER' framework (Sample, Phenomenon of Interest, Design, Evaluation, Research type) (Cooke et al., 2012). Studies were included where: i) the sample were university students (UG or PG level); ii) the main phenomenon of interest was practice/s which aimed to improve performance on oral assessment; iii) any type of experimental design or qualitative design is used; iv) the evaluation provided an indication as to the extent to which the practice aided students; v) the research was empirical (as opposed to review articles or theoretical papers). In addition, only articles written in English were included in the review. See Table 1 for further detail.

Identification and screening of studies

The PRISMA diagram (Figure 1) provides details regarding the databases searched for the review. Searches were conducted in November 2024. Filters were applied such as English language and peer reviewed. The date range was 2015 to present day. This was due to the previous systematic literature review by Van Ginkel et al. (2015). Despite the existence of this review, scoping searches revealed there to be a sufficient number of studies conducted between 2015 and 2024 to conduct a further systematic literature review to add to the body of literature.

Searches across the four databases yielded 941 results. Of these, 256 were duplicates and, as such, were removed. A further 621 were removed following screening of titles and, where necessary, abstracts. Sixty-four articles were screened against the inclusion criteria by the lead author. A sub-set of articles (15%) were reviewed against the criteria by the two other authors in order to reduce the potential for bias in selection. Consensus was agreed through discussion between authors in the small number of cases where a disagreement occurred. The reference lists of the 22 articles which remained after this screening process were manually searched, this yielded a further three studies, two of which met inclusion criteria. The final number of articles for inclusion in the review was 24.

All studies were quality assessed using the Mixed Methods Appraisal Tool (MMAT) (Hong et al., 2018) by the lead author. The decision was taken for all papers to be included in the review.

Table 1. Study selection criteria based on the SPIDER framework.

Category	Inclusion criteria	Exclusion criteria
Sample (S)	University students (UG or PG level)	Studies with non-university student samples (e.g. professionals, school students)
Phenomenon of interest (PI)	Practices/interventions aimed at improving performance on oral assessment	Studies focusing on other types of assessment (e.g. written exams, coursework) where data specifically on oral assessment cannot be extracted. Papers focusing on the PhD Viva voce will be excluded
Design (D)	Experimental or qualitative designs	Theoretical papers, opinion pieces, or non-empirical studies
Evaluation (E)	Provides an indication of how the practice/interventions improved performance	No clear evaluation of impact
Research type (R)	Empirical research (qualitative or quantitative)	Literature reviews, meta-analyses, conceptual papers

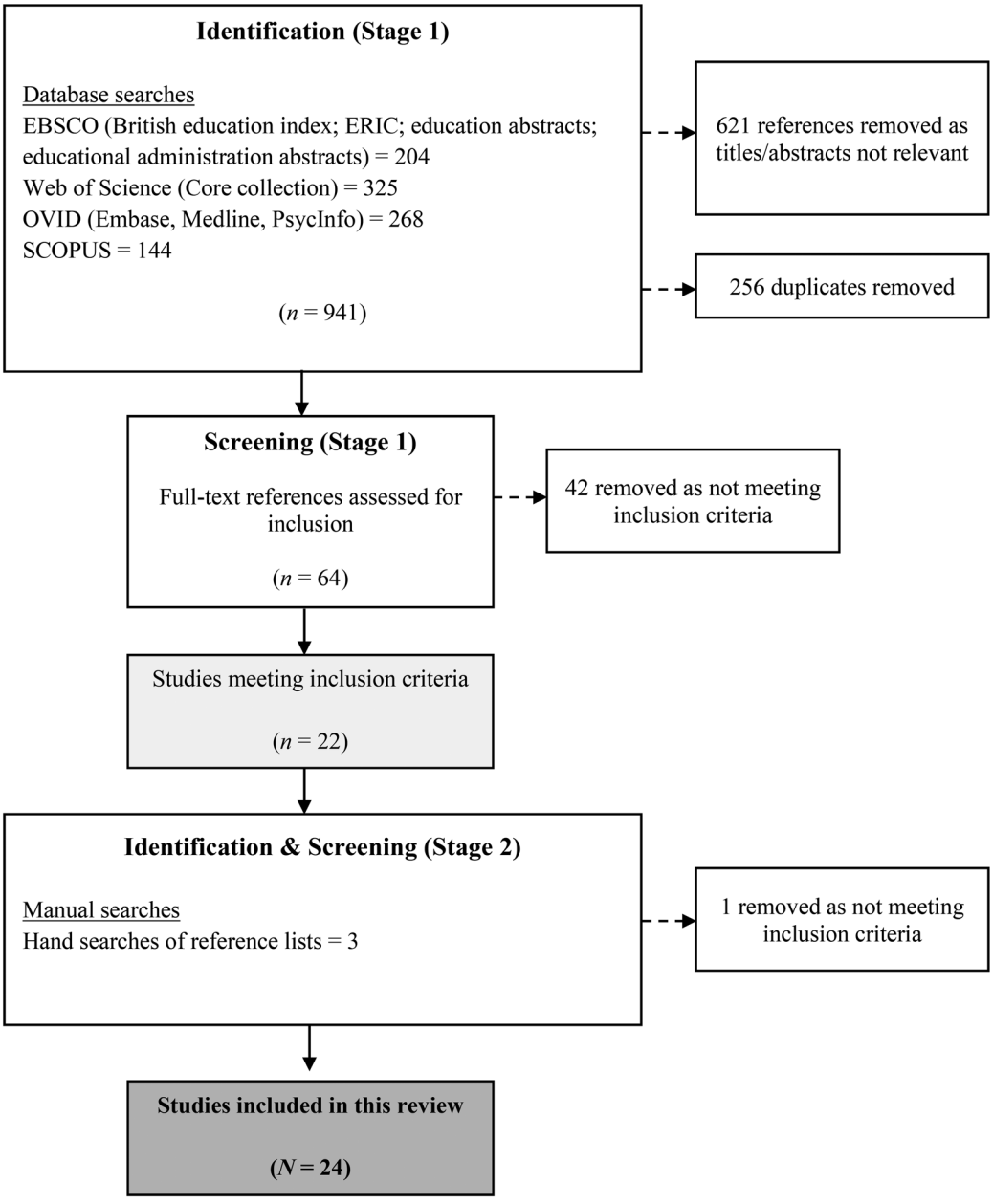

Figure 1. PRISMA diagram.

Results

Study characteristics

Country

Many of studies included in this review were conducted in the USA ($n = 10$, 41.7%), followed by the Netherlands ($n = 4$, 16.7%), and Australia ($n = 4$, 16.7%). Additional studies originated from Spain ($n = 2$, 8.3%), the UK ($n = 1$, 4.2%), Italy ($n = 1$, 4.2%), Canada ($n = 1$, 4.2%), and Sweden ($n = 1$, 4.2%).

Discipline

The studies covered a range of disciplines, with the some focusing on engineering ($n=4$, 16.7%), followed by medical sciences ($n=3$, 12.5%), chemistry ($n=2$, 8.3%), and communications ($n=2$, 8.3%). The remaining studies included students from a range of disciplines, such as forensic psychology, business, pharmacy, biology, and life sciences.

Research design

The studies employed diverse research designs, with experimental designs being the most prevalent ($n=11$, 45.8%), followed by quasi-experimental designs ($n=5$, 20.8%), and qualitative studies ($n=3$, 12.5%). Additional approaches included survey designs ($n=2$, 8.3%), mixed methods ($n=2$, 8.3%), and one correlational study ($n=1$, 4.2%).

Participants

As specified in the inclusion criteria, all participants were university students. Where specified, participants were predominantly undergraduate students ($n=16$, 66.7%), with the balance being postgraduate students ($n=6$, 25.0%). Sample sizes varied widely, ranging from 11 to 789 participants, with two studies not reporting the number of participants.

Intervention

A variety of interventions were described, with 11 studies examining feedback in various forms (e.g. from peers, teachers; 45.8%), six involving the use of virtual reality (VR; 25.0%), six focusing on self-reflection (25.0%), and five utilising activities such as games, structured modules, or novel techniques to improve communication and presentation skills (20.8%). Please note the total exceeds 24 because some studies implemented multiple intervention strategies, such as combining feedback with self-reflection or virtual reality techniques.

Narrative synthesis

Given the inclusion of both qualitative, quantitative and mixed-method studies, a narrative synthesis was identified as the most suitable approach for analysing the data. This synthesis was conducted in line with the guidance provided by Popay et al. (2006) for synthesising findings in systematic reviews. The data are organised under headings that reflect the key patterns and areas of focus identified across the studies. Where particularly relevant, key statistical findings are highlighted to illustrate the effectiveness of specific interventions, while more detailed quantitative data and study characteristics can be found in the Supplementary Table.

Technology

Eight studies explored the use of technology-based methods to enhance oral presentation skills, primarily through Virtual Reality (VR) and video-based feedback.

Virtual reality (VR) interventions

VR has been used as an immersive tool for public speaking practice. Belboukhaddaoui and Van Ginkel (2019) found that both immediate and delayed computer-generated feedback in VR improved presentation performance with no significant differences between feedback timings ($p>0.05$). Similarly, Boetje et al. (2020) showed that VR-based practice significantly improved oral presentation performance ($M=42.82$ to $M=49.85$; $p<0.001$), with greater gains observed after multiple sessions. Davis et al. (2020) reported that VR practice resulted in significantly higher

public speaking anxiety ($p = 0.05$) compared to face-to-face practice, though final performance scores did not differ significantly ($p = 0.39$). According to the study authors, this increased anxiety may have been due to the heightened sense of presence in VR, making the audience feel more real and amplifying apprehension. Some participants also perceived negative audience feedback, and difficulties reading notes in VR added to their stress, highlighting the need for careful implementation (Davis et al., 2020).

Pradhananga et al. (2022) also found that VR, when combined with other activities, significantly improved students' presentation skills and confidence ($p < 0.001$), with minority students reporting greater confidence gains. Additionally, Van Ginkel et al. (2019) highlighted the effectiveness of VR-based practice and feedback methods, demonstrating improvements in students' presentation performance and engagement.

Video recordings

Video recordings were used in several studies, either to support student reflection on their performance or as a tool for staff to assess presentations remotely. Galindo et al. (2020) demonstrated that lecturer feedback on recorded presentations led to significant improvements in organisation, text delivery, and body language ($p < 0.05$). Additionally, Oliveira et al. (2021) utilised video recordings to help students engage in self-reflection, discussed further under the self-reflection theme.

Online communication tools

McBain et al. (2016) explored the use of online oral communication assessment tasks (OOCATs), finding that these tools positively influenced students' perceptions of their oral communication skills and contributed to improvements in presentation performance. As McBain et al. (2016) is the only identified study on this specific approach, further research would be needed to establish broader conclusions about its effectiveness.

Peer and instructor feedback approaches

Nine studies emphasised the importance of feedback from instructors and peers in improving oral presentation skills.

Instructor feedback

Instructor feedback was found to be a critical factor in improving presentation quality. Van Ginkel et al. (2017) compared four feedback conditions—self, peer, teacher, and peer guided by teacher—and found that teacher feedback led to the highest performance scores ($t = 0.47$, $p = 0.02$). Similarly, Gunderson et al. (2021) reported that faculty evaluations resulted in the most substantial improvements, with faculty scores increasing by 6.60 points ($p < 0.01$) following structured feedback.

Galindo et al. (2020) demonstrated that lecturer feedback on recorded presentations significantly improved organisation, text delivery, and body language ($p < 0.05$), though verbal language remained unchanged. Additionally, Sox et al. (2018) compared a web-based feedback module with in-person faculty-led feedback, finding that both methods significantly improved presentation performance compared to the control group, with in-person feedback yielding slightly greater improvements.

Peer feedback

Studies examining peer feedback found it to be effective but variable in long-term retention. Murillo-Zamorano and Montanero (2018) demonstrated that students in the peer feedback

condition showed a 10% improvement in presentation scores, though this effect diminished after one month.

Dickson et al. (2019) explored both the impact of giving and receiving peer feedback, finding that students who actively participated as peer reviewers—in addition to receiving feedback—scored significantly higher ($p < 0.001$) compared to those who did not engage in peer review activities. Nikolic et al. (2018) and Ritchie (2016) further supported the value of peer feedback, with both studies highlighting improvements in students' critical reflection and oral communication skills through peer-assessment activities.

However, Olsson Jers and Wärnsby (2018) noted that while peer feedback prompted reflection, it was not always effectively applied in subsequent presentations, particularly when feedback was affirmatory rather than constructive.

Structured Training and repeated practice

Six studies focused on formal training exercises and repeated practice to improve presentation confidence and delivery.

Mock oral examinations

Fischer et al. (2016) and Higgins et al. (2016) examined the impact of mock oral examinations, structured assessments designed to prepare students for professional certification or competency examinations. In Fischer et al. (2016), participation in mock oral examinations led to a significant improvement in performance, with first-time pass rates increasing from 76.0% to 100% ($p = 0.025$). Higgins et al. (2016) found that 77.1% of students reported the mock oral examination as very or extremely helpful in preparing for certification examinations.

Gradual exposure and enactive mastery

Cavanagh et al. (2019) uniquely explored the impact of enactive mastery, vicarious experience, verbal persuasion, and physiological arousal techniques within a structured business communication course. This combination of strategies resulted in a significant increase in students' self-efficacy regarding presenting ($M = 74.69$ to $M = 91.62$; $p < 0.001$).

Novel techniques in structured training

Applebee et al. (2018) introduced a three-minute slide activity aimed at boosting students' presentation confidence without over-reliance on slides. Over 90% of students and alumni perceived the activity as beneficial, agreeing or strongly agreeing that it helped improve their future presentation skills. Stout (2020) required students to create TED Talk-style presentations, which reportedly enhanced student engagement, creativity, and oral communication skills, as reflected in improved presentation scores. Nash et al. (2016) implemented public speaking skill development exercises focused on reducing anxiety and improving confidence through structured exposure and feedback. Significant improvements were observed in students' confidence with managing nerves, maintaining eye contact, and using gestures effectively ($p < 0.05$), although anxiety remained a dominant emotion for many during public speaking tasks.

Self-reflection and metacognitive strategies

Four studies focused on developing self-awareness and reflection to enhance students' presentation performance.

Self-Assessment and reflective writing

Tsingos-Lucas et al. (2017) found that reflective writing scores significantly predicted performance on oral examinations ($p < 0.01$). Serafini and Chittaro (2024) used a VR oral examination simulator, with students who engaged in self-assessment scoring significantly higher ($p = 0.03$) in clarity, timing, and content organisation. Oliveira et al. (2021) and Nikolic et al. (2018) highlighted the role of self-reflection through video recordings and formative self-assessment tasks, leading to greater critical awareness of presentation skills and performance improvement.

Discussion

Summary of main findings

This systematic review identified several key interventions that enhance student performance in oral assessments across diverse academic disciplines. Consistent with prior findings (Van Ginkel et al., 2015), feedback emerged as a critical facilitator, with both instructor-led (e.g. Gunderson et al., 2021) and peer-based feedback (e.g. Murillo-Zamorano & Montanero, 2018) contributing significantly to performance improvements. Structured practice opportunities, such as mock oral examinations, demonstrated positive effects on student preparedness and confidence (Fischer et al., 2016; Higgins et al., 2016). Self-reflective strategies, including video-based self-assessment and reflective writing, were also associated with enhanced performance by fostering metacognitive awareness (Tsingos-Lucas et al., 2017; Oliveira et al., 2021).

Notably, the integration of technology, particularly virtual reality (VR), represents a significant advancement since earlier reviews. VR interventions showed promise in improving presentation skills, although mixed outcomes were observed regarding their impact on student anxiety (Boetje et al., 2020; Davis et al., 2020).

Interpretation of findings

Feedback benefits performance on oral assessments across educational contexts

The effectiveness of feedback can be attributed to its role in providing specific, actionable insights that guide students' learning processes (Hattie & Timperley, 2007). Instructor feedback, in particular, demonstrated a strong impact on performance outcomes (Van Ginkel et al., 2017), likely due to the perceived authority and expertise of educators in shaping student development. Peer feedback, while effective, showed variability in its long-term retention, suggesting the need for structured peer review protocols to maximise its benefits (Murillo-Zamorano & Montanero, 2018). In addition, the findings suggest that structured peer review protocols emphasising constructive rather than merely affirmative feedback are necessary to maximise its benefits (Murillo-Zamorano & Montanero, 2018).

The finding regarding the effectiveness of mock oral assessments was consistent with literature on preparing for the PhD viva examination (Lantsoght, 2018). Mock orals in the studies reviewed (Fischer et al., 2016; Higgins et al., 2016) were typically part of coursework or professional training assessments, distinct from doctoral viva examinations. These mock assessments provided opportunities for targeted feedback, refinement of responses, and the development of critical thinking under examination-like conditions. This aligns with viva preparation literature, which highlights the value of practice in reducing uncertainty, enhancing self-efficacy, and supporting students to perform to the best of their ability (Lantsoght, 2018).

These findings extend those of Van Ginkel et al. (2015), whose review identified feedback—particularly in terms of its intensity, timing, and source—as a key design principle for developing oral presentation competence. The present review reinforces this and provides updated empirical support across a broader range of oral assessment contexts.

VR is a promising tool for oral assessments, but implementation needs careful consideration

Van Ginkel et al. (2015) did not explore the use of digital technologies in oral assessments, as such, the current review extends their review by considering how emerging tools—particularly Virtual Reality (VR)—can enhance or hinder oral presentation competence. The findings related to the effectiveness of VR in the learning of oral presentation skills in the current review can be viewed under the broader banner of evolving educational landscape where technological tools are increasingly incorporated into assessment practices. For example, Technology-Enhanced Learning (TEL) is increasingly prevalent in higher education, encompassing digital tools like virtual learning environments, simulations, and online platforms to create flexible, engaging learning experiences (Sümer & Vaněček, 2024). Within TEL, Virtual Reality (VR) offers immersive, interactive environments that simulate real-world scenarios, providing experiential learning opportunities beyond traditional classroom settings. In nursing, for example, VR has been used to safely develop clinical skills (Liu et al., 2023). Research shows VR can enhance theoretical knowledge, practical skills, and learner satisfaction (Yu & Xu, 2022; Liu et al., 2023). However, VR's effectiveness varies based on implementation, highlighting the need for educators to understand how to use it effectively (Sümer & Vaněček, 2024). When integrated thoughtfully, VR can create dynamic, engaging learning experiences tailored to diverse student needs. However, if not carefully designed, VR may increase anxiety, particularly in high-stakes scenarios, reinforcing the need for thoughtful integration and appropriate support mechanisms.

These findings build on those of Miskam and Saidalvi (2020), who reviewed the use of video technology to improve undergraduate students' oral presentation skills and found consistent benefits in areas such as content, fluency, organisation, and self-awareness. The current review broadens this technological lens to include VR.

Self-reflection enhances oral assessment performance by engaging critical thinking

Van Ginkel et al. (2015) identified self-assessment—particularly through tools like video review and portfolios—as a key design principle for developing oral presentation competence. The present review builds on this by highlighting the broader role of self-reflection in fostering critical thinking and metacognitive awareness, and by considering the additional value of rubrics in supporting transparent and consistent self-evaluation within oral assessment contexts. More specifically, this review found that self-reflective practices enhanced students' ability to critically evaluate their performance and identify areas for improvement. This aligns with metacognitive theories that emphasise the importance of self-evaluation in the learning process (Stanton et al., 2021). Furthermore, this finding is consistent with broader educational literature, which highlights reflection as a vital learning strategy that fosters critical thinking and deepens learning (Boud et al., 1985).

It is noteworthy that many studies in this review employed rubrics for both formative and summative oral assessments. Rubrics have been found to help students understand expectations and performance standards, enabling them to better prepare (Ragupathi & Lee, 2020). Using standard rubrics, even for diverse presentation topics, promotes consistency and fairness in grading. This is perhaps particularly important in oral assessments, where the lack of anonymity could introduce examiner bias. While, to the authors' knowledge, there are, as yet, no studies providing evidence of bias directly linked to non-anonymised assessments, research shows that a significant number of students perceive fairness as a concern in non-anonymised assessments (Borkin, 2020). Standard rubrics may help mitigate actual and perceived biases by ensuring clear, consistent marking criteria.

Strengths and limitations

A key strength of this review is the inclusion of studies spanning a wide range of disciplines and intervention types, offering a comprehensive understanding of factors influencing oral

assessment performance. This diversity enriches the applicability of findings across different educational contexts.

However, several limitations warrant consideration. The review was restricted to English-language publications, potentially excluding relevant studies from non-English-speaking contexts and introducing language bias. Additionally, while the inclusion of diverse interventions and disciplines enhances breadth, it may limit the generalisability of findings to specific educational settings. Some studies also featured small sample sizes (e.g. Dickson et al., 2019), which may reduce the robustness of individual results and affect the reliability of conclusions.

Implications for educators

The findings offer practical implications for educators seeking to improve student outcomes in oral assessments. In the ever-evolving landscape of higher education, educators are increasingly tasked with designing assessments that are resilient to the capabilities of generative artificial intelligence (GAI) (Moorhouse et al., 2023). Oral assessments have gained attention as a viable approach to mitigate AI-related challenges, as they require spontaneous, real-time responses that are less susceptible to AI interference (Eachempati et al., 2025). This shift necessitates a re-evaluation of traditional assessment methods to ensure academic integrity and the accurate evaluation of student learning.

However, implementing oral assessments introduces considerations regarding student anxiety, as some learners find these formats to be particularly stress-inducing (Huxham et al., 2012). To support students in performing to the best of their abilities, it is essential to incorporate strategies that alleviate anxiety and build confidence. The results of this review would suggest that providing structured practice sessions and delivering timely, constructive feedback are effective methods to enhance students' oral competencies and potentially to reduce anxiety.

Educators are encouraged to adapt these evidence-based strategies to their specific contexts, experimenting with different approaches to identify what works best for their student populations. Additionally, fostering a culture that values feedback and reflection can create supportive learning environments that promote continuous improvement in oral communication skills.

Future research directions

Future research should aim to address the limitations identified in this review. Including studies published in languages other than English could provide a more comprehensive understanding of global educational practices. Longitudinal studies are needed to examine the sustained impact of interventions, particularly technology-based approaches, on oral assessment performance.

Further exploration of the mechanisms through which feedback and reflective practices influence learning outcomes could also enhance the development of effective assessment strategies.

Disclosure statement

No potential conflict of interest was reported by the authors.

Notes on contributors

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Sam Cruchley is a Professor of Materials Science and Engineering Education within the School of Metallurgy and Materials and the current Deputy Director of Education for the College of Engineering and Physical Sciences. In these roles he has a significant teaching and leadership roles within STEM education and conducts research in both materials science and engineering and pedagogy. His currently research interests lie in understanding the mechanical property, oxidation and corrosion interactions in high performance metallic alloys, as well as on authentic, innovative & synoptic teaching and assessment practices, with a view to improving student engagement, employability and widening participation.

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